🧘 Adaptive Yoga Recommendation System

The **Adaptive Yoga Recommendation System** is an AI-powered wellness assistant designed to help users enhance their yoga practice through personalized guidance and intelligent insights.

🧠 What It Does

This system analyzes the user's **journal input**—either written or spoken—and combines it with their **personal health data** and **pose accuracy trends** to recommend suitable yoga poses. The recommendations are tailored to support recovery, improve form, and promote consistency in practice.

🔍 Key Features

- 📓 **Journal Input (Text or Voice):**
  Users can write or record a short journal about their current mood, fatigue, or pain. Voice entries are transcribed using the Whisper model (via `faster-whisper`).

- 🧑‍⚕️ **User Profile Awareness:**
  The system adapts based on personal details like age, gender, injury history, and fitness goals to recommend safe and effective poses.

- 📊 **Weekly Progress Tracking:**
  Tracks daily pose accuracy, recurring posture errors, and fatigue trends across sessions using visual line and bar charts.

- 💡 **Intelligent Pose Suggestions:**
  Based on issues detected from the journal (e.g., "back pain", "fatigue", "stress", "balance problems"), it suggests 2–4 beginner-friendly poses, prioritizing safety and user needs.

- 🌐 **Integrated Learning Support:**
  Each recommended pose includes:

  - A **tutorial video** link for guidance.
  - A **Wikipedia snippet** for background information.
  - Live **DuckDuckGo search results** with resources for deeper learning.

- 📋 **Summary Report:**
  Each recommendation includes a motivational quote, practice tips, and mentions injury-specific modifications if applicable.

🧰 Built With

- Python (Streamlit, SQLite3, pandas)
- Faster-Whisper for transcription
- Web scraping (DuckDuckGo & Wikipedia)
- Matplotlib (for optional word clouds/visuals)

Setup Guide -

This guide helps you run the Yoga Recommendation App on your own machine. It supports both text and voice journaling, processes user performance data, and gives personalized yoga suggestions using NLP and pose matching.

📁 Project Folder Structure

```
yoga_project/
│
├── app.py                  # Streamlit frontend
├── yoga_agent.py           # Recommendation logic, NLP, Whisper
├── requirements.txt        # All required Python packages
├── yoga_recommendation.db (optional) # DB file (auto-created)
```

🛠️ 1. Prerequisites

✅ Install Anaconda

If not already installed, download and install:
🔗 https://www.anaconda.com/products/distribution

🧪 2. Create and Activate Virtual Environment

Option 1: Using Anaconda Prompt

bash Copy code

```
conda create -n yoga_env python=3.11
conda activate yoga_env
```

📦 3. Install Dependencies

Step 1: Navigate to the project folder

bash Copy code

```
cd C:\Users\<your_username>\Desktop\yoga_project
```

Step 2: Install packages from `requirements.txt`

bash Copy code

```
pip install -r requirements.txt
```

🎧 4. Set Up FFmpeg (for Voice Input)

Whisper requires `ffmpeg` to process audio:

1. Download from: ffmpeg builds

2. Extract it to `C:\ffmpeg`
3. Add the `bin` folder to your PATH:
    - Search for **"Environment Variables"** in Start Menu
    - Edit **Path** under system variables
    - Add: `C:\ffmpeg\bin`
4. Restart your Anaconda Prompt.

▶️ 5. Run the App

Inside your project folder:

```bash
streamlit run app.py
```

The app will open in your browser.

🧠 Features

- 📝 Text or voice journaling (Whisper transcription)
- 📊 Weekly progress charts (accuracy + fatigue)
- 🧘 Personalized yoga pose recommendations
- 🎥 Tutorial links + Wikipedia info
- 🧠 NLP-based issue detection (back pain, fatigue, posture, etc.)

📂 Optional: Create the Initial Database

If you're running for the first time and need to create the SQLite DB:

You can call `create_and_populate_db()` function in a one-time script or notebook to seed demo users and sessions.

👩‍💻 Developer Notes

- All user voice input is transcribed via `faster-whisper`.
- You can modify the yoga knowledge base or logic in `yoga_agent.py`.
- Weekly report is generated from actual session and journal logs.

❤️ Contact

If you face issues running the app, feel free to open an issue or reach out!